STD 590-85 0732290 0011754 1
E-26-06

Steel Line Blanks

API STANDARD 590
FIRST EDITION, MARCH 1985

American Petroleum Institute
1220 L Street, Northwest
Washington, D.C. 20005



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Steel Line Blanks

Refining Department

API STANDARD 590
FIRST EDITION, MARCH 1985

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FOREWORD

This standard is a purchase specification for steel line blanks intended for installation between flanges conforming to ANSI B16.5 dimensions.

Suggested revisions are invited and should be submitted to the director of the Refining Department, American Petroleum Institute, 1220 L Street, N.W., Washington, D.C. 20005.

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Figure
1—Steel Line Blanks 1

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NOTES TO PURCHASER

1. If the purchaser needs a steel line blank that deviates from this standard, the deviating requirements shall be stated in the purchase order.
2. If no exceptions are to be taken to this standard, the purchase order need only make reference to API Standard 590 and specify the items marked with an asterisk (*) in the following list. Other items listed are options that may also be specified if required.

*a. Size (see 1.1).
*b. Class (see 2.2).
*c. Facing (see 3.3).
*d. Style (see 1.2).
*e. Plate material (see Section 5).
f. Inside diameter (see 4.2).
g. Lifting or handling device (see 3.1.2).

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Steel Line Blanks

SECTION 1—GENERAL

1.1 Scope

This standard provides dimensions for operating line blanks (or "blinds") in sizes NPS ½ through NPS 24 for installation between ANSI B16.5 flanges in the 150, 300, 600, 900, 1500, and 2500 pound pressure classes. The dimensions are suitable for blanks made of ten different steel plate materials. Line blanks used only for pressure testing are not part of this standard.

1.2 Definitions

1.2.1 A *Figure 8 blank* (also called a spectacle blank) is a pressure retaining plate with one solid end and one open end connected with a web or tie bar (see Figure 1).

1.2.2 A *paddle blank* is similar to the solid end of a figure 8 blank (with a handle) and is generally used in conjunction with a paddle spacer in large sizes (see Figure 1).

1.2.3 A *paddle spacer* is similar to the open end of a figure 8 blank (with a handle) and is generally used in conjunction with a paddle blank (see Figure 1).

1.3 Referenced Publications

The latest edition or revision of the following standards or specifications shall, to the extent specified in this standard, form a part of this standard.

ANSI[1]/ASME[2]

B16.5 *Pipe Flanges and Flanged Fittings*
B16.20 *Ring-Joint Gaskets and Grooves for Steel Pipe Flanges*
B16.34 *Valves, Flanged and Buttwelding End*
B31.3 *Chemical Plant and Petroleum Refinery Piping*

ANSI

B46.1 *Surface Texture*

ASTM[3]

A 204 *Pressure Vessel Plates, Alloy Steel, Molybdenum*
A 240 *Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels*
A 387 *Pressure Vessel Plates, Alloy Steel, Chromium-Molybdenum*
A 515 *Pressure Vessel Plates, Carbon Steel, for Intermediate- and Higher-Temperature Service*
A 516 *Pressure Vessel Plates, Carbon Steel, for Moderate- and Lower-Temperature Service*

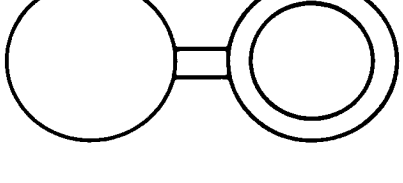
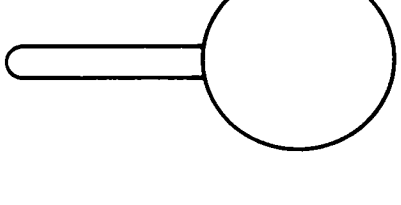
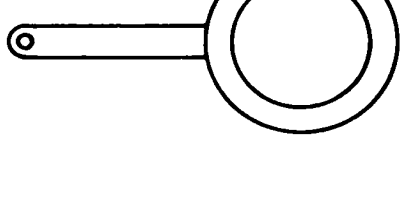


FIGURE 8 BLANK



PADDLE BLANK



PADDLE SPACER

Figure 1—Steel Line Blanks

[1] American National Standards Institute, 1430 Broadway, New York, New York 10018.
[2] American Society of Mechanical Engineers, 345 East 47th Street, New York, New York 10017.
[3] American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.

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SECTION 2—RATINGS

2.1 Pressure-Temperature Ratings

Pressure-temperature ratings are those listed in ANSI B16.5 and B16.34 standard class for the material group corresponding to the plate materials.

2.2 ANSI Class

Line blanks covered by this standard are for the following ANSI classes: 150, 300, 600, 900, 1500, and 2500.

SECTION 3—DESIGN

3.1 Handle

3.1.1 The handle or web (tie-bar) may be integral or attached to the line blank or spacer by welding. Any weld shall be capable of supporting the weight of the blank or spacer in all orientations and shall meet the visual inspection requirements of ANSI/ASME B31.3.

3.1.2 When a lifting or handling device is deemed necessary by the purchaser, the purchaser shall provide the necessary details to the supplier.

3.2 Edge Preparation

Flame, plasma, saw cutting, or press punching are acceptable methods for forming the inside and outside diameters of line blanks. Surfaces shall be free of projections that would interfere with gasket seating.

3.3 Facing

3.3.1 RAISED FACE JOINT BLANKS

A raised face may be specified at the option of the purchaser. The gasket seating surface finish and dimensions for raised face line blanks shall be in accordance with ANSI B16.5. The height of the raised faces shall be in addition to the thicknesses listed in Tables 2 through 19.

3.3.2 FEMALE RING-JOINT BLANKS

Female ring-joint grooves shall be shaped for octagonal rings with the groove side wall surface finish not rougher than 63 microinches AARH. Roughness shall be determined in accordance with ANSI B46.1.

3.3.3 MALE RING-JOINT BLANKS

The gasket shape (ring) for male ring-joint blanks shall be oval unless an octagonal ring is specified in the purchase order. Surface finish on the sealing surface shall not be rougher than 63 microinches AARH. Roughness shall be determined in accordance with ANSI B46.1.

SECTION 4—DIMENSIONS

4.1 General

4.1.1 Dimensions shall be in accordance with Tables 2 through 19 of this standard.

4.1.2 Dimensional tolerances shall be established by the manufacturer but shall not exceed ANSI B16.5 tolerances for similar aspects.

4.1.3 Line blank thickness is designed in accordance with ANSI/ASME B31.3.

4.2 Openings

4.2.1 For NPS ½, ¾, and 1 in all raised face classes, the inside diameter is equal to standard weight weld neck flange bore.

4.2.2 For NPS 1¼ and larger in Class 150 and 300 raised face, the inside diameter is equal to pipe outside diameter.

4.2.3 For NPS 1¼ and larger in Class 600 and 900 raised face, the inside diameter is equal to Schedule 10S weld neck flange bore.

4.2.4 For Class 1500 raised face, the inside diameter is equal to Schedule 40 weld neck flange bore.

4.2.5 For Class 2500 raised face, the inside diameter is

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equal to Schedule 40 through NPS 6, Schedule 60 for NPS 8 and 10, and Schedule 80 for NPS 12.

4.2.6 For all ring joint classes, the inside diameter is equal to pipe outside diameter.

4.2.7 Dimensions are based upon concentric installation on spiral wound gaskets with inner rings as required by API Standard 601 and conform to the maximum permitted bore of ANSI B16.5 weld neck flanges described in Table C-1 of API Standard 601, Sixth Edition.

SECTION 5—MATERIALS

Table 1—Blank Construction Materials

Material Group	ASTM Plate Material	Maximum Operating Temperature, F
1.1	A 515-70, A 516-70	800
1.7	A 204-C	850
1.9	A 387-11 CL2	1100 (Note 1)
1.10	A 387-22 CL2	1100 (Note 1)
1.12	A 387-5 CL2	1100 (Note 1)
2.1	A 240-304	1000 (Notes 1,2,3)
2.2	A 240-316	1000 (Notes 1,2,3)
2.4	A 240-321	1000 (Notes 1,2,3)
2.5	A 240-347	1000 (Notes 1,2,3)

NOTES:
1. For Class 150, the maximum temperature permitted is 1000 F (540 C).
2. At temperatures over 1000 F (540 C), use the material only if it is heat treated to a temperature of at least 1900 F (1040 C) and quenched in water or rapidly cooled by other means and if the carbon content is 0.04 percent or higher.
3. Thicknesses listed in Tables 2 through 19 are suitable to 1000 F (540 C). Use at higher temperatures is permitted, however, the purchaser is required to ascertain suitability.

5.1 Table 1 lists the materials that are acceptable for the manufacture of line blanks conforming to the dimensions in Tables 2 through 19.

5.2 The thicknesses in Tables 2 through 19 for blanks made of the materials listed in Table 1 are adequate for all pressure-temperature combinations within the specified flange pressure class up to the maximum temperatures listed in Table 1.

SECTION 6—MARKING

6.1 General

Line blanks conforming to this standard shall be marked as follows:

1. API-590.
2. Nominal pipe size (NPS).
3. ANSI class.
4. Material, specification, and grade.
5. Manufacturer's name or trademark.

Example:
API-590
3-300
ASTM 515 Gr 70
xyz co.

6.2 Marking Method

The marking shall be applied by steel stamping on the web (tie bar) or handle. Where space is limited, marking may be stamped on the blind portion of blanks (but not on the gasket seating surface) or on the circumference.

SECTION 7—VISUAL IDENTIFICATION

Paddle spacer handles shall have one ½ inch (13 millimeter) minimum diameter hole located near the outer end. This hole serves as a remote visual indicator.

CAUTION: Paddle blanks shall not be supplied with indicator or bolt holes.

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SECTION 8—SHIPMENT

8.1 Coating

All machined surfaces (except those on austenitic materials) shall be coated with an easily removable rust preventive.

8.2 Shipment Protection

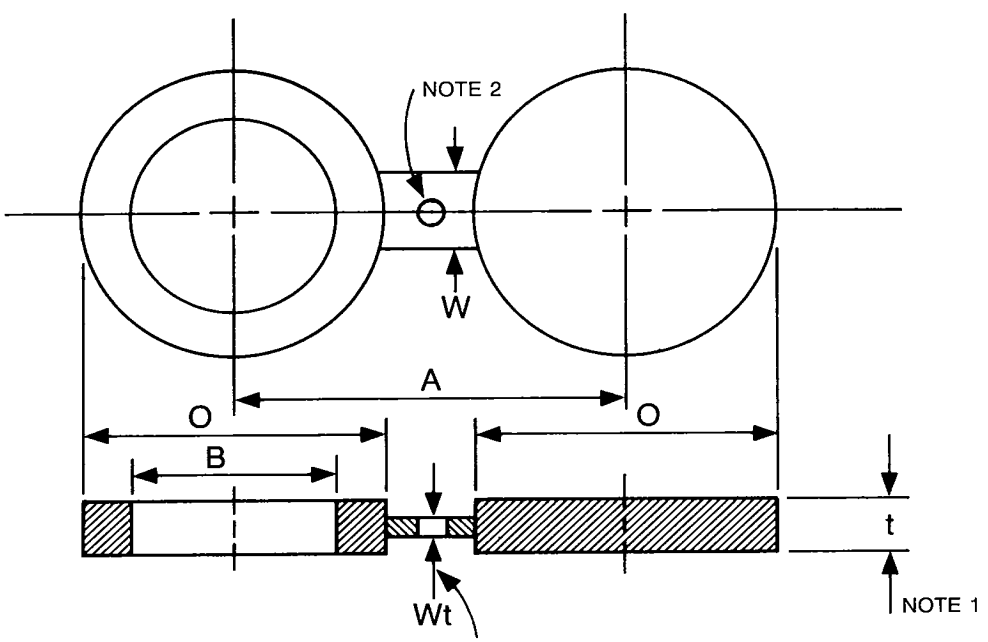
Machined surfaces shall be protected from damage during storage and shipping with a metal, wood, or plastic covering.

SECTION 9—TESTING

Line blanks are not required to be hydrostatically tested.

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Table 2—Dimensions of Class 150 Raised Face Figure 8 Blanks



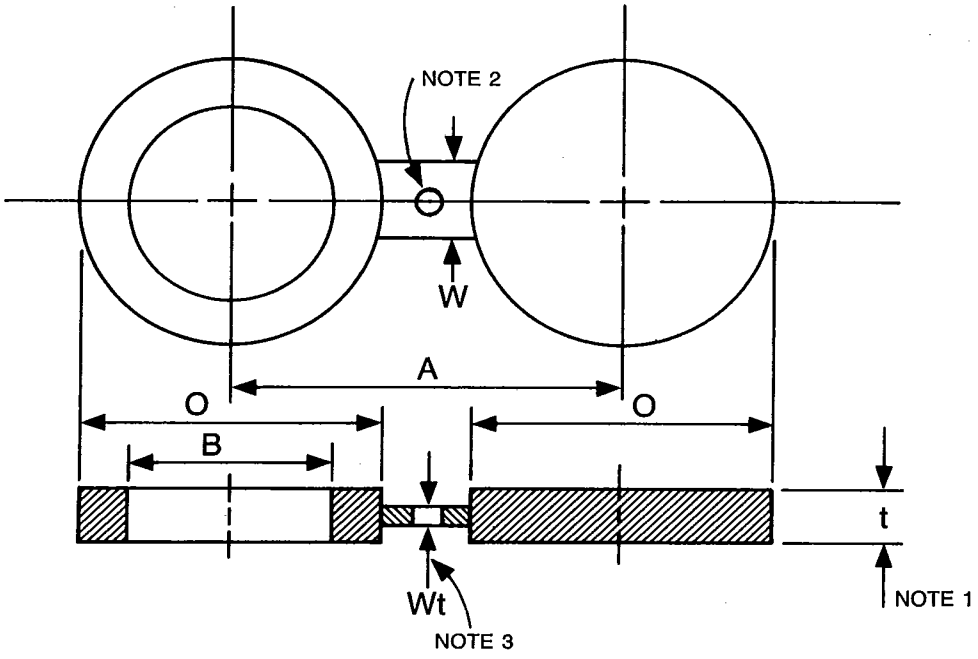
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.62	16	1.75	44	2.38	60	0.12	3	1.50	38
¾	0.82	21	2.12	54	2.75	70	0.12	3	1.50	38
1	1.05	27	2.50	64	3.12	79	0.12	3	1.50	38
1 ¼	1.66	42	2.88	73	3.50	89	0.25	6	1.50	38
1 ½	1.90	48	3.25	83	3.88	99	0.25	6	1.50	38
2	2.38	60	4.00	102	4.75	121	0.25	6	2.00	51
2 ½	2.88	73	4.75	121	5.50	140	0.25	6	2.00	51
3	3.50	89	5.25	133	6.00	152	0.25	6	2.50	64
3 ½	4.00	102	6.25	159	7.00	178	0.38	10	2.50	64
4	4.50	114	6.75	171	7.50	190	0.38	10	2.50	64
5	5.56	141	7.62	194	8.50	216	0.38	10	3.00	76
6	6.62	168	8.62	219	9.50	241	0.50	13	3.00	76
8	8.62	219	10.88	276	11.75	298	0.50	13	3.00	76
10	10.75	273	13.25	337	14.25	362	0.62	16	4.00	102
12	12.75	324	16.00	406	17.00	432	0.75	19	4.00	102
14	14.00	356	17.62	448	18.75	476	0.75	19	4.25	108
16	16.00	406	20.12	511	21.25	540	0.88	22	4.25	108
18	18.00	457	21.50	546	22.75	578	1.00	25	4.50	114
20	20.00	508	23.75	603	25.00	635	1.12	28	4.75	121
24	24.00	610	28.12	714	29.50	749	1.25	32	5.50	140

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.

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Table 3—Dimensions of Class 300 Raised Face Figure 8 Blanks



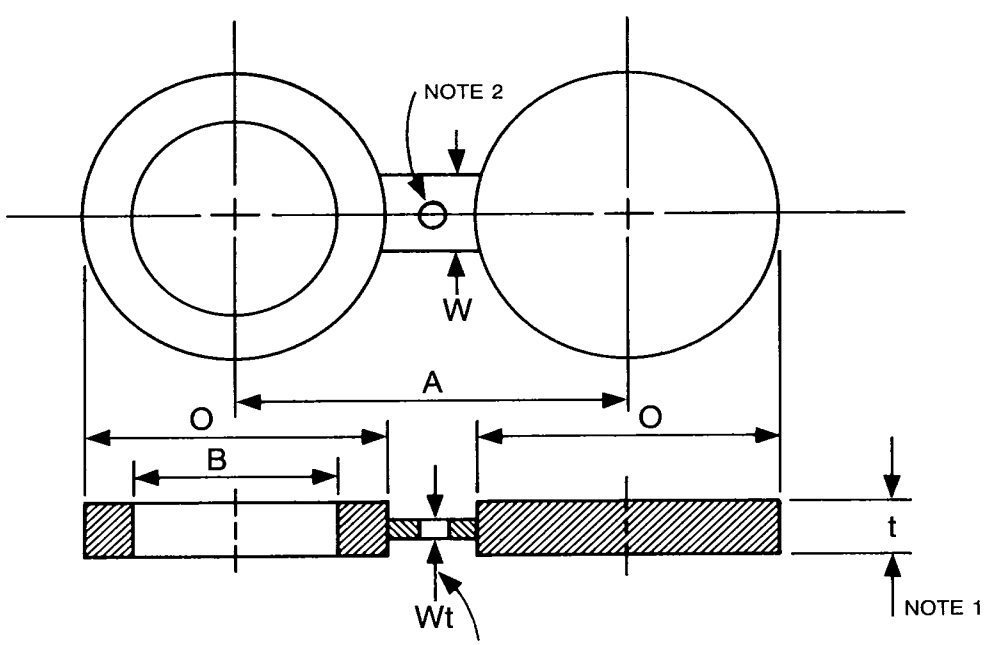
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.62	16	2.00	51	2.62	67	0.25	6	1.50	38
¾	0.82	21	2.50	64	3.25	83	0.25	6	1.50	38
1	1.05	27	2.75	70	3.50	89	0.25	6	1.50	38
1 ¼	1.66	42	3.12	79	3.88	99	0.25	6	1.50	38
1 ½	1.90	48	3.62	92	4.50	114	0.25	6	2.00	51
2	2.38	60	4.25	108	5.00	127	0.38	10	2.00	51
2 ½	2.88	73	5.00	127	5.88	149	0.38	10	2.50	64
3	3.50	89	5.75	146	6.62	168	0.38	10	2.50	64
3 ½	4.00	102	6.38	162	7.25	184	0.50	13	2.50	64
4	4.50	114	7.00	178	7.88	200	0.50	13	2.50	64
5	5.56	141	8.38	213	9.25	235	0.62	16	3.00	76
6	6.62	168	9.75	248	10.62	270	0.62	16	3.00	76
8	8.62	219	12.00	305	13.00	330	0.88	22	3.50	89
10	10.75	273	14.12	359	15.25	387	1.00	25	4.00	102
12	12.75	324	16.50	419	17.75	451	1.12	28	4.00	102
14	14.00	356	19.00	483	20.25	514	1.25	32	4.75	121
16	16.00	406	21.12	537	22.50	572	1.50	38	4.88	124
18	18.00	457	23.38	594	24.75	629	1.62	41	4.50	114
20	20.00	508	25.62	651	27.00	686	1.75	44	4.75	121
24	24.00	610	30.38	772	32.00	813	2.00	51	5.50	140

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.

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Table 4—Dimensions of Class 600 Raised Face Figure 8 Blanks



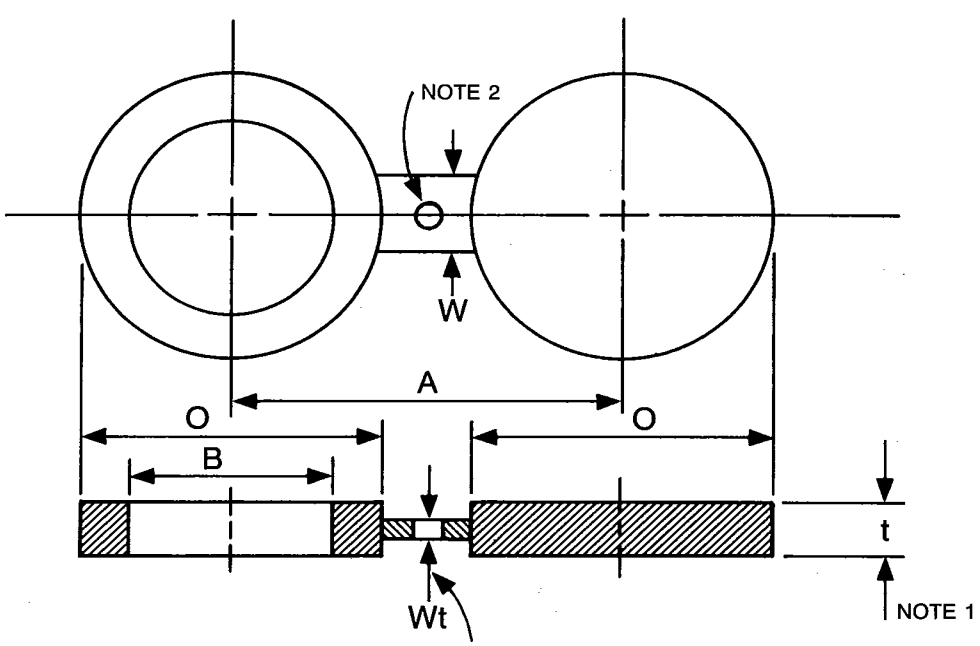
NPS	Inside Diameter B		Outside Diameter O		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.62	16	2.00	51	2.62	67	0.25	6	1.50	38
¾	0.82	21	2.50	64	3.25	83	0.25	6	1.50	38
1	1.05	27	2.75	70	3.50	89	0.25	6	2.25	57
1 ¼	1.44	37	3.12	79	3.88	99	0.38	10	2.25	57
1 ½	1.68	43	3.62	92	4.50	114	0.38	10	2.62	67
2	2.16	55	4.25	108	5.00	127	0.38	10	2.25	57
2 ½	2.64	67	5.00	127	5.88	149	0.50	13	2.62	67
3	3.26	83	5.75	146	6.62	168	0.50	13	2.62	67
3 ½	3.76	96	6.25	159	7.25	184	0.62	16	3.00	76
4	4.26	108	7.50	191	8.50	216	0.62	16	3.00	76
5	5.30	135	9.38	238	10.50	267	0.75	19	3.38	86
6	6.36	162	10.38	264	11.50	292	0.88	22	3.38	86
8	8.33	212	12.50	318	13.75	349	1.12	28	3.75	95
10	10.42	265	15.62	397	17.00	432	1.38	35	4.12	105
12	12.39	315	17.88	454	19.25	489	1.62	41	4.12	105
14	13.62	346	19.25	489	20.75	527	1.75	44	4.50	114
16	15.62	397	22.12	562	23.75	603	2.00	51	4.88	124
18	17.62	448	24.00	610	25.75	654	2.12	54	5.25	133
20	19.56	497	26.75	679	28.50	724	2.50	64	5.25	133
24	23.50	597	31.00	787	33.00	838	2.88	73	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.

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Table 5—Dimensions of Class 900 Raised Face Figure 8 Blanks



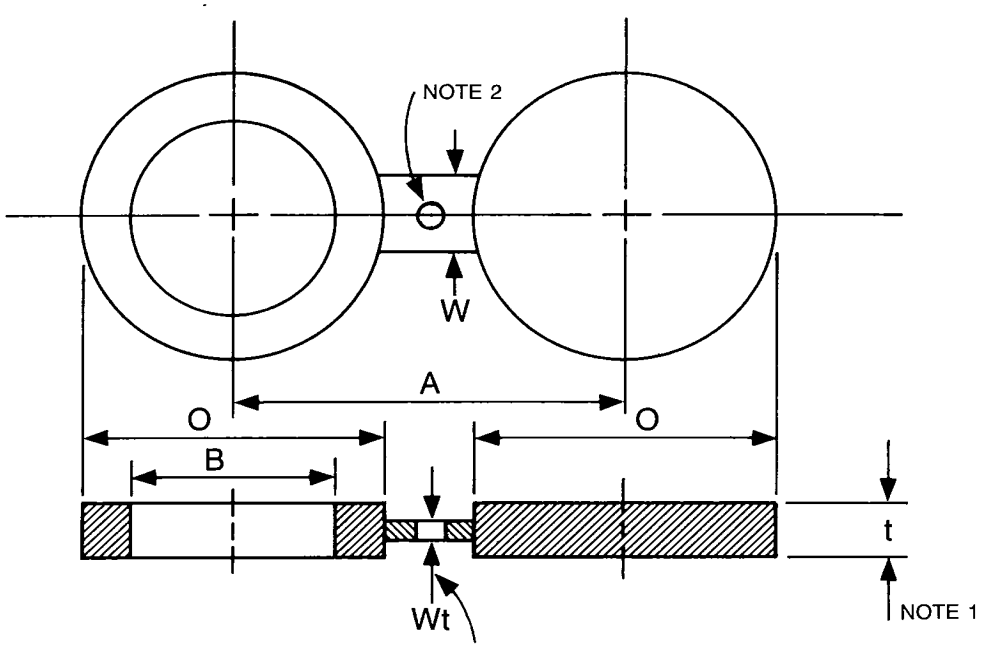
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.62	16	2.38	60	3.25	83	0.25	6	1.50	38
¾	0.82	21	2.62	67	3.50	89	0.25	6	1.62	41
1	1.05	27	3.00	76	4.00	102	0.25	6	2.25	57
1 ¼	1.44	37	3.38	86	4.38	111	0.38	10	2.25	57
1 ½	1.68	43	3.75	95	4.88	124	0.38	10	2.62	67
2	2.16	55	5.50	140	6.50	165	0.50	13	2.25	57
2 ½	2.64	67	6.38	162	7.50	190	0.50	13	2.62	67
3	3.26	83	6.50	165	7.50	190	0.62	16	2.62	67
4	4.26	108	8.00	203	9.25	235	0.75	19	3.00	76
5	5.30	135	9.62	244	11.00	279	0.88	22	3.38	86
6	6.36	162	11.25	286	12.50	318	1.00	25	3.38	86
8	8.33	212	14.00	356	15.50	394	1.38	35	3.75	95
10	10.42	265	17.00	432	18.50	470	1.62	41	4.12	105
12	12.39	315	19.50	495	21.00	533	1.88	48	4.12	105
14	13.62	346	20.38	518	22.00	559	2.12	54	4.50	114
16	15.62	397	22.50	572	24.25	616	2.38	60	4.88	124
18	17.62	448	25.00	635	27.00	686	2.62	67	5.25	133
20	19.56	497	27.38	695	29.50	749	2.88	73	5.25	133
24	23.50	597	32.88	835	35.50	902	3.50	89	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.

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Table 6—Dimensions of Class 1500 Raised Face Figure 8 Blanks



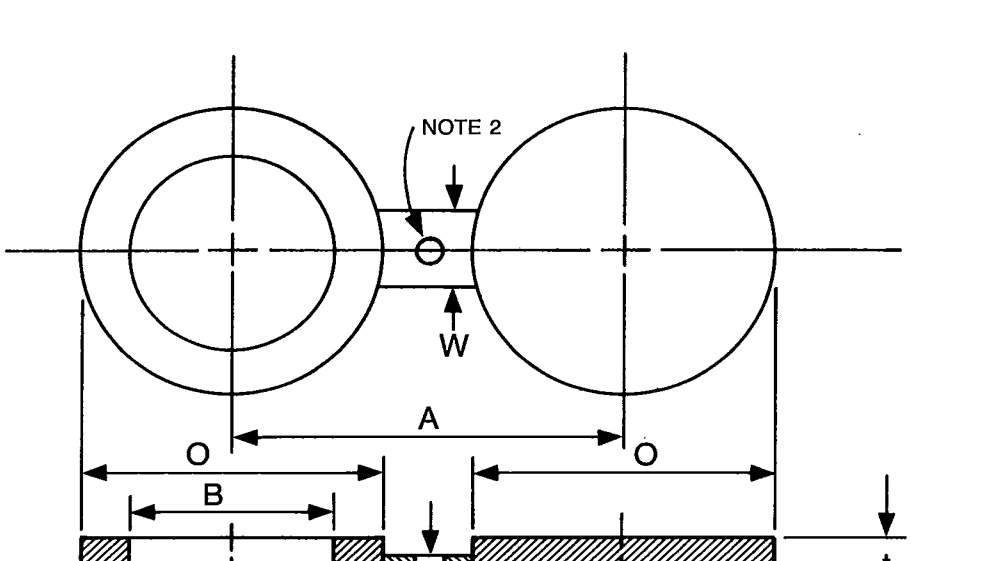
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.62	16	2.38	60	3.25	83	0.25	6	1.50	38
¾	0.82	21	2.62	67	3.50	89	0.38	10	1.62	41
1	1.05	27	3.00	76	4.00	102	0.38	10	2.50	64
1 ¼	1.38	35	3.38	86	4.38	111	0.38	10	2.50	64
1 ½	1.61	41	3.75	95	4.88	124	0.50	13	2.75	70
2	2.07	53	5.50	140	6.50	165	0.50	13	2.75	70
2 ½	2.47	63	6.38	162	7.50	190	0.62	16	3.00	76
3	3.07	78	6.75	171	8.00	203	0.75	19	3.00	76
4	4.03	102	8.12	206	9.50	241	0.88	22	3.50	89
5	5.05	128	9.88	251	11.50	292	1.12	28	3.50	89
6	6.06	154	11.00	279	12.50	318	1.38	35	3.50	89
8	7.98	203	13.75	349	15.50	394	1.62	41	4.00	102
10	10.02	255	17.00	432	19.00	483	2.00	51	4.50	114
12	11.94	303	20.38	518	22.50	572	2.38	60	4.50	114
14	13.12	333	22.62	575	25.00	635	2.62	67	5.00	127
16	15.00	381	25.12	638	27.75	705	3.00	76	5.25	133
18	16.88	429	27.62	702	30.50	775	3.38	86	5.75	146
20	18.81	478	29.62	752	32.75	832	3.75	95	6.00	152
24	22.62	575	35.38	899	39.00	991	4.38	111	7.00	178

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.

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Table 7—Dimensions of Class 2500 Raised Face Figure 8 Blanks



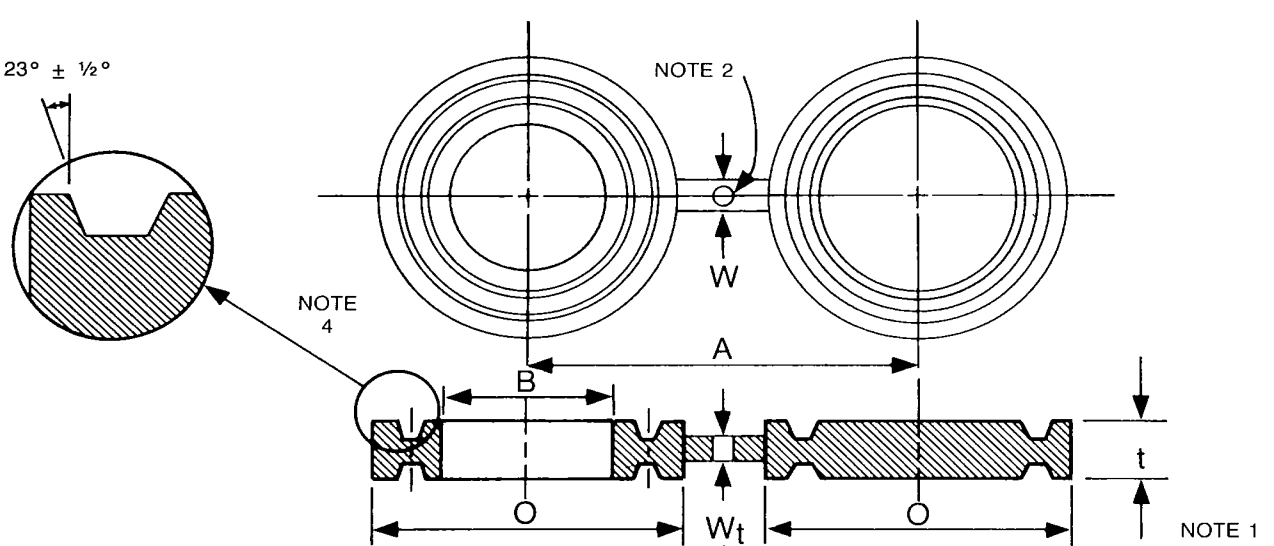
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.62	16	2.62	67	3.50	89	0.38	10	1.50	38
¾	0.82	21	2.88	73	3.75	95	0.38	10	1.62	41
1	1.05	27	3.25	83	4.25	108	0.38	10	2.50	64
1 ¼	1.38	35	4.00	102	5.12	130	0.50	13	2.50	64
1 ½	1.61	41	4.50	114	5.75	146	0.62	16	2.75	70
2	2.07	53	5.62	143	6.75	171	0.62	16	2.75	70
2 ½	2.47	63	6.50	165	7.75	197	0.75	19	3.00	76
3	3.07	78	7.62	194	9.00	229	0.88	22	3.00	76
4	4.03	102	9.12	232	10.75	273	1.12	28	3.50	89
5	5.05	128	10.88	276	12.75	324	1.38	35	3.50	89
6	6.06	154	12.38	314	14.50	368	1.62	41	3.50	89
8	7.81	198	15.12	384	17.25	438	2.12	54	4.00	102
10	9.75	248	18.62	473	21.25	540	2.62	67	4.50	114
12	11.37	289	21.50	546	24.38	619	3.12	79	4.50	114

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.

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Table 8—Dimensions of Class 150 Female Ring-Joint Facing Figure 8 Blanks



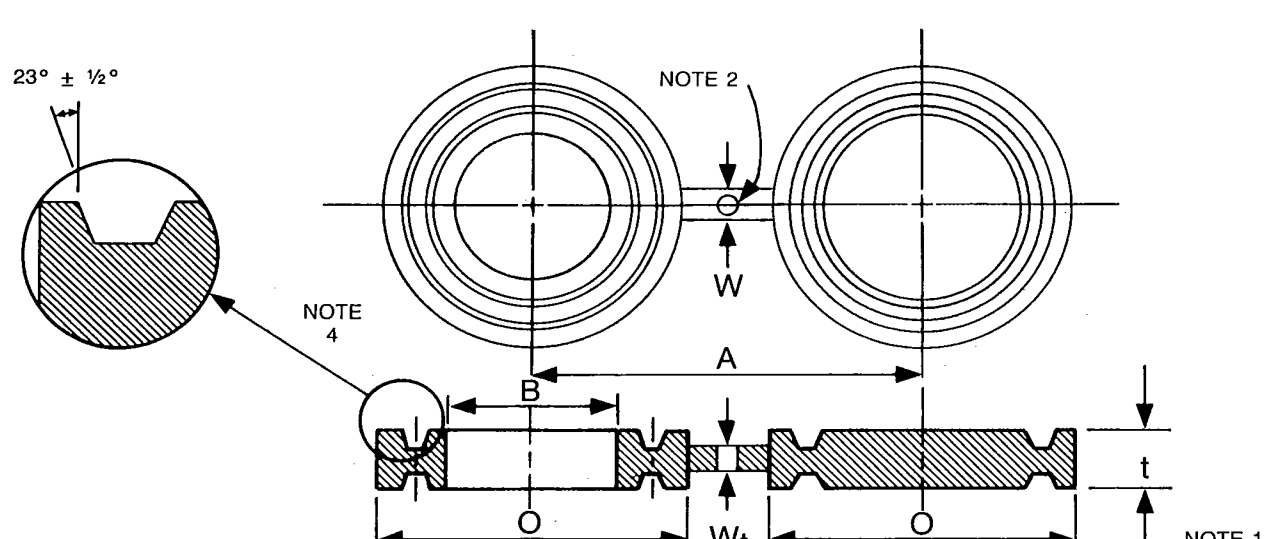
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
1	1.32	34	2.50	64	3.12	79	0.75	19	2.00	51
1 ¼	1.66	42	2.88	73	3.50	89	0.75	19	2.00	51
1 ½	1.90	48	3.25	83	3.88	99	0.75	19	2.25	57
2	2.38	60	4.00	102	4.75	121	0.75	19	2.25	57
2 ½	2.88	73	4.75	121	5.50	140	0.88	22	2.25	57
3	3.50	89	5.25	133	6.00	152	0.88	22	2.25	57
3 ½	4.00	102	6.06	154	7.00	178	0.88	22	2.50	64
4	4.50	114	6.75	171	7.50	190	0.88	22	2.50	64
5	5.56	141	7.62	194	8.50	216	1.00	25	2.75	70
6	6.62	168	8.62	219	9.50	241	1.00	25	3.25	83
8	8.62	219	10.75	273	11.75	298	1.12	28	3.75	95
10	10.75	273	13.00	330	14.25	362	1.25	32	4.00	102
12	12.75	324	16.00	406	17.00	432	1.38	35	4.75	121
14	14.00	356	16.75	425	18.75	476	1.38	35	5.00	127
16	16.00	406	19.00	483	21.25	540	1.50	38	5.00	127
18	18.00	457	21.50	546	22.75	578	1.62	41	5.00	127
20	20.00	508	23.50	597	25.00	635	1.62	41	5.00	127
24	24.00	610	28.00	711	29.50	749	1.88	48	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Female ring joint groove dimensions shall be designed for octagonal rings in accordance with ANSI B16.5.

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Table 9—Dimensions of Class 300 Female Ring-Joint Facing Figure 8 Blanks



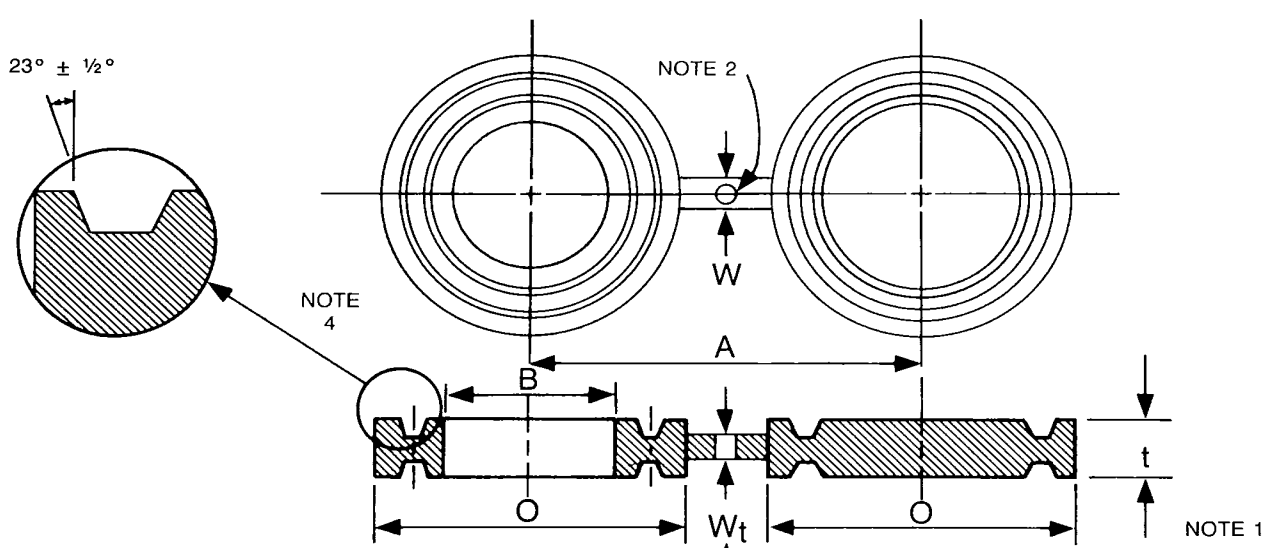
NPS	Inside Diameter B		Outside Diameter O		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.00	51	2.62	67	0.62	16	1.50	38
¾	1.05	27	2.50	64	3.25	83	0.75	19	1.75	44
1	1.32	34	2.75	70	3.50	89	0.75	19	2.00	51
1 ¼	1.66	42	3.12	79	3.88	99	0.88	22	2.00	51
1 ½	1.90	48	3.56	90	4.50	114	0.88	22	2.25	57
2	2.38	60	4.25	108	5.00	127	1.00	25	2.25	57
2 ½	2.88	73	5.00	127	5.88	149	1.12	28	2.25	57
3	3.50	89	5.75	146	6.62	168	1.12	28	2.25	57
3 ½	4.00	102	6.25	159	7.25	184	1.12	28	2.50	64
4	4.50	114	6.88	175	7.88	200	1.25	32	2.50	64
5	5.56	141	8.25	210	9.25	235	1.38	35	2.75	70
6	6.62	168	9.50	241	10.62	270	1.38	35	3.25	83
8	8.62	219	11.88	302	13.00	330	1.62	41	3.75	95
10	10.75	273	14.00	356	15.25	387	1.75	44	4.00	102
12	12.75	324	16.25	413	17.75	451	2.00	51	4.75	121
14	14.00	356	18.00	457	20.25	514	2.12	54	5.00	127
16	16.00	406	20.00	508	22.50	572	2.25	57	5.00	127
18	18.00	457	22.62	575	24.75	629	2.38	60	5.00	127
20	20.00	508	25.00	635	27.00	686	2.75	70	5.00	127
24	24.00	610	29.50	749	32.00	813	3.12	79	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Female ring joint groove dimensions shall be designed for octagonal rings in accordance with ANSI B16.5.

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Table 10—Dimensions of Class 600 Female Ring-Joint Facing Figure 8 Blanks



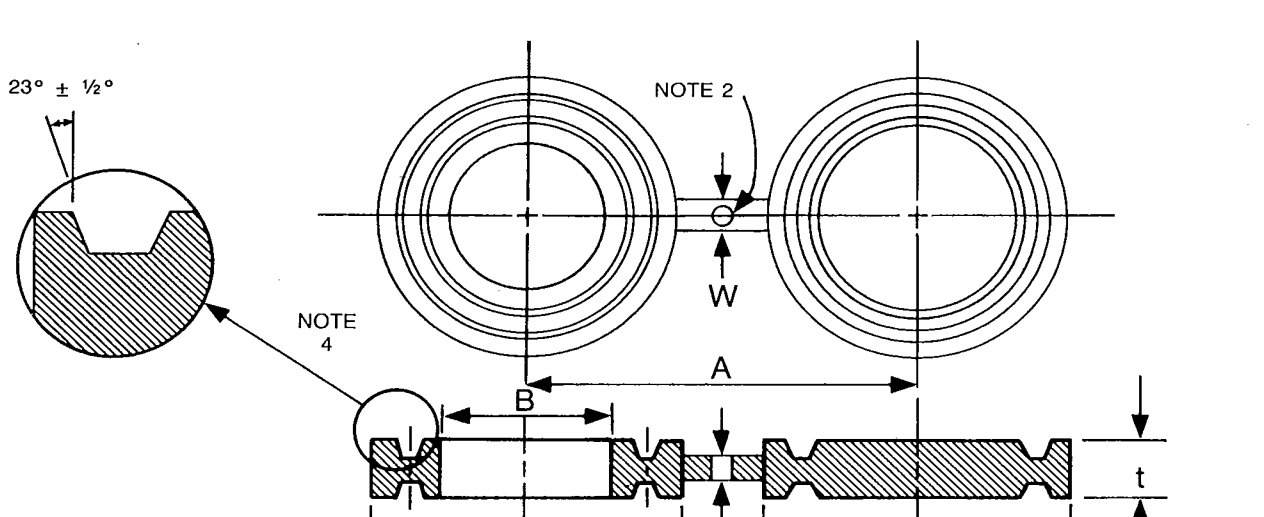
	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
NPS	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.00	51	2.62	67	0.75	19	1.50	38
¾	1.05	27	2.50	64	3.25	83	0.88	22	1.75	44
1	1.32	34	2.75	70	3.50	89	0.88	22	2.00	51
1 ¼	1.66	42	3.12	79	3.88	99	0.88	22	2.00	51
1 ½	1.90	48	3.56	90	4.50	114	0.88	22	2.25	57
2	2.38	60	4.25	108	5.00	127	1.12	28	2.25	57
2 ½	2.88	73	5.00	127	5.88	149	1.25	32	2.25	57
3	3.50	89	5.75	146	6.62	168	1.25	32	2.25	57
3 ½	4.00	102	6.25	159	7.25	184	1.38	35	2.50	64
4	4.50	114	6.88	175	8.50	216	1.38	35	2.50	64
5	5.56	141	8.25	210	10.50	267	1.50	38	2.75	70
6	6.62	168	9.50	241	11.50	292	1.75	44	3.25	83
8	8.62	219	11.88	302	13.75	349	2.00	51	3.75	95
10	10.75	273	14.00	356	17.00	432	2.25	57	4.00	102
12	12.75	324	16.25	413	19.25	489	2.50	64	4.75	121
14	14.00	356	18.00	457	20.75	527	2.62	67	5.00	127
16	16.00	406	20.00	508	23.75	603	2.88	73	5.00	127
18	18.00	457	22.62	575	25.75	654	3.12	79	5.00	127
20	20.00	508	25.00	635	28.50	724	3.50	89	5.00	127
24	24.00	610	29.50	749	33.00	838	4.12	105	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Female ring joint groove dimensions shall be designed for octagonal rings in accordance with ANSI B16.5.

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Table 11—Dimensions of Class 900 Female Ring-Joint Facing Figure 8 Blanks



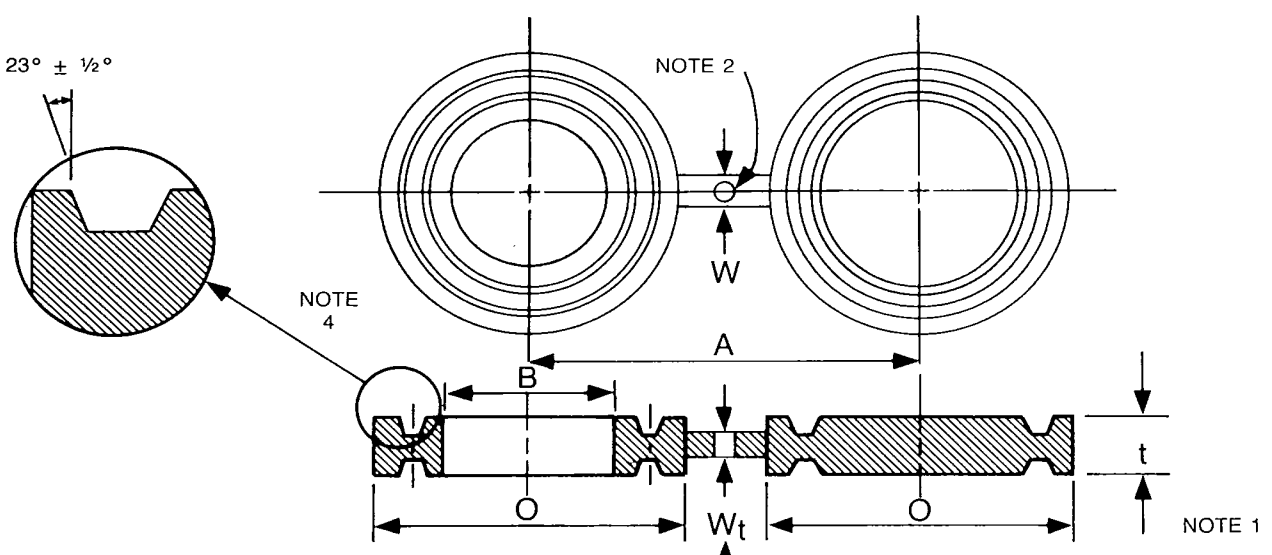
NPS	Inside Diameter B		Outside Diameter O		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.38	60	3.25	83	0.88	22	1.50	38
¾	1.05	27	2.62	67	3.50	89	0.88	22	1.75	44
1	1.32	34	2.81	71	4.00	102	0.88	22	2.00	51
1 ¼	1.66	42	3.19	81	4.38	111	1.00	25	2.00	51
1 ½	1.90	48	3.62	92	4.88	124	1.00	25	2.50	64
2	2.38	60	4.88	124	6.50	165	1.25	32	2.00	51
2 ½	2.88	73	5.38	137	7.50	190	1.38	35	2.62	67
3	3.50	89	6.12	156	7.50	190	1.38	35	2.62	67
4	4.50	114	7.12	181	9.25	235	1.62	41	2.88	73
5	5.56	141	8.50	216	11.00	279	1.75	44	2.88	73
6	6.62	168	9.50	241	12.50	318	1.88	48	2.88	73
8	8.62	219	12.12	308	15.50	394	2.25	57	3.12	79
10	10.75	273	14.25	362	18.50	470	2.50	64	4.75	121
12	12.75	324	16.50	419	21.00	533	2.88	73	4.75	121
14	14.00	356	18.38	467	22.00	559	3.25	83	4.75	121
16	16.00	406	20.62	524	24.25	616	3.62	92	5.00	127
18	18.00	457	23.38	594	27.00	686	4.00	102	5.25	133
20	20.00	508	25.50	648	29.50	749	4.38	111	5.00	127
24	24.00	610	30.38	772	35.50	902	5.25	133	5.50	140

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Female ring joint groove dimensions shall be designed for octagonal rings in accordance with ANSI B16.5.

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Table 12—Dimensions of Class 1500 Female Ring-Joint Facing Figure 8 Blanks



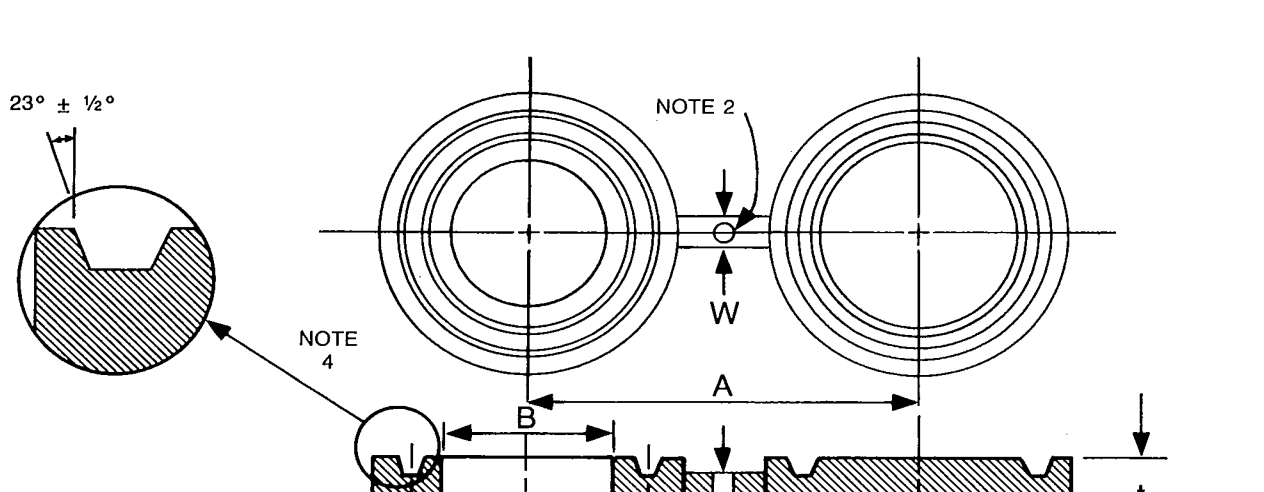
NPS	Inside Diameter B		Outside Diameter O		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.38	60	3.25	83	0.88	22	1.50	38
¾	1.05	27	2.62	67	3.50	89	1.00	25	1.75	44
1	1.32	34	2.81	71	4.00	102	1.00	25	2.12	54
1 ¼	1.66	42	3.19	81	4.38	111	1.00	25	2.12	54
1 ½	1.90	48	3.62	92	4.88	124	1.12	28	2.25	57
2	2.38	60	4.88	124	6.50	165	1.38	35	2.12	54
2 ½	2.88	73	5.38	137	7.50	190	1.50	38	2.25	57
3	3.50	89	6.62	168	8.00	203	1.75	44	2.88	73
4	4.50	114	7.62	194	9.50	241	1.88	48	3.00	76
5	5.56	141	9.00	229	11.50	292	2.12	54	3.00	76
6	6.62	168	9.75	248	12.50	318	2.38	60	3.12	79
8	8.62	219	12.50	318	15.50	394	2.88	73	3.38	86
10	10.75	273	14.62	371	19.00	483	3.25	83	5.25	133
12	12.75	324	17.25	438	22.50	572	4.00	102	5.25	133
14	14.00	356	19.25	489	25.00	635	4.38	111	5.50	140
16	16.00	406	21.50	546	27.75	705	4.88	124	5.75	146
18	18.00	457	24.12	613	30.50	775	5.25	133	6.00	152
20	20.00	508	26.50	673	32.75	832	5.62	143	6.50	165
24	24.00	610	31.25	794	39.00	991	6.62	168	7.00	178

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Female ring joint groove dimensions shall be designed for octagonal rings in accordance with ANSI B16.5.

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Table 13—Dimensions of Class 2500 Female Ring-Joint Facing Figure 8 Blanks



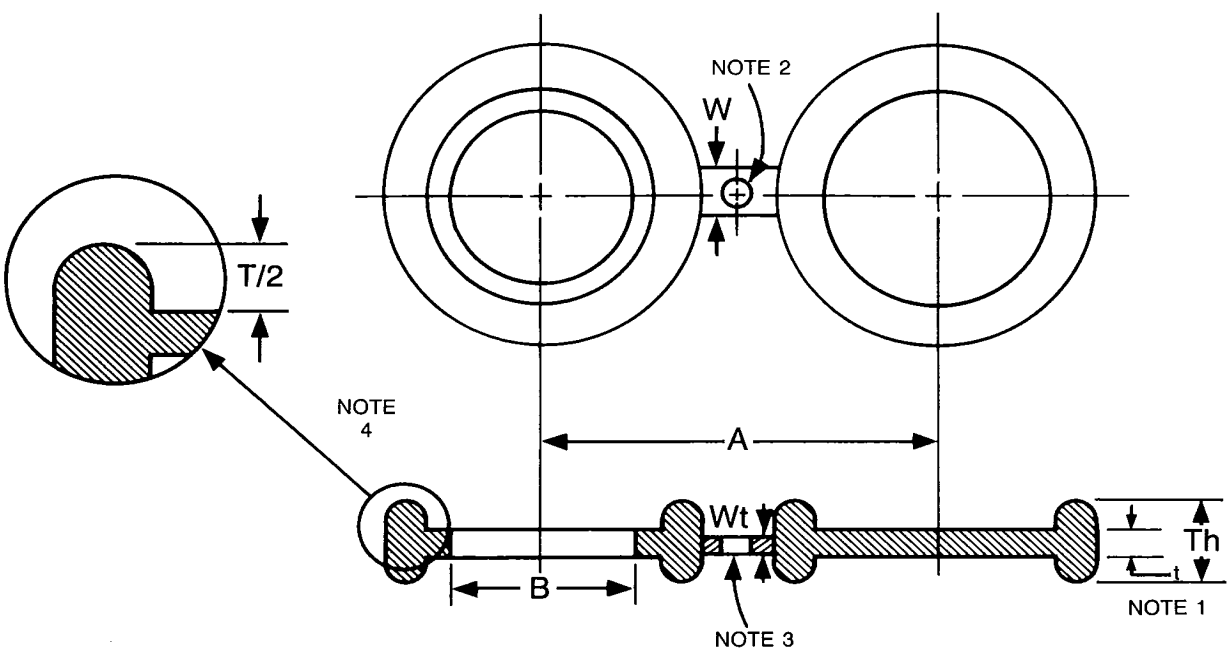
NPS	Inside Diameter *B*		Outside Diameter *O*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.56	65	3.50	89	1.00	25	1.50	38
¾	1.05	27	2.88	73	3.75	95	1.12	28	1.75	44
1	1.32	34	3.25	83	4.25	108	1.12	28	2.12	54
1 ¼	1.66	42	4.00	102	5.12	130	1.38	35	2.12	54
1 ½	1.90	48	4.50	114	5.75	146	1.50	38	2.38	60
2	2.38	60	5.25	133	6.75	171	1.62	41	2.25	57
2 ½	2.88	73	5.88	149	7.75	197	1.88	48	2.38	60
3	3.50	89	6.62	168	9.00	229	2.00	51	3.00	76
4	4.50	114	8.00	203	10.75	273	2.50	64	3.25	83
5	5.56	141	9.50	241	12.75	324	2.88	73	3.50	89
6	6.62	168	11.00	279	14.50	368	3.25	83	3.75	95
8	8.62	219	13.38	340	17.25	438	3.88	99	3.75	95
10	10.75	273	16.75	425	21.25	540	4.62	117	3.58	91
12	12.75	324	19.50	495	24.38	619	5.25	133	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Female ring joint groove dimensions shall be designed for octagonal rings in accordance with ANSI B16.5.

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Table 14—Dimensions of Class 150 Male Oval Ring-Joint Facing Figure 8 Blanks



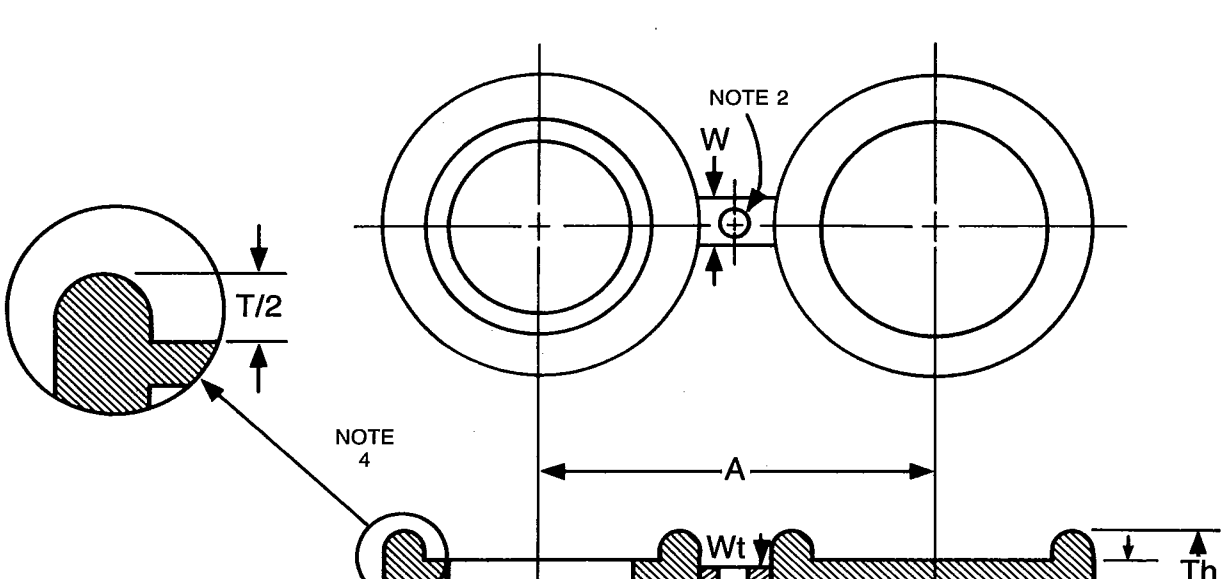
NPS	Inside Diameter B		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
1	1.32	34	3.12	79	0.25	6	2.00	51
1 ¼	1.66	42	3.50	89	0.25	6	2.00	51
1 ½	1.90	48	3.88	99	0.25	6	2.25	57
2	2.38	60	4.75	121	0.25	6	2.25	57
2 ½	2.88	73	5.50	140	0.38	10	2.25	57
3	3.50	89	6.00	152	0.38	10	2.25	57
3 ½	4.00	102	7.00	178	0.38	10	2.50	64
4	4.50	114	7.50	190	0.38	10	2.50	64
5	5.56	141	8.50	216	0.50	13	3.00	76
6	6.62	168	9.50	241	0.50	13	3.25	83
8	8.62	219	11.75	298	0.62	16	3.75	95
10	10.75	273	14.25	362	0.75	19	4.00	102
12	12.75	324	17.00	432	0.88	22	4.75	121
14	14.00	356	18.75	476	0.88	22	5.00	127
16	16.00	406	21.25	540	1.00	25	5.00	127
18	18.00	457	22.75	578	1.12	28	5.00	127
20	20.00	508	25.00	635	1.12	28	5.00	127
24	24.00	610	29.50	749	1.38	35	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Oval ring joint dimensions shall be in accordance with ANSI B16.20 except $Th = T+t$ where *T* is the ring height specified in B16.20.

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Table 15—Dimensions of Class 300 Male Oval Ring-Joint Facing Figure 8 Blanks



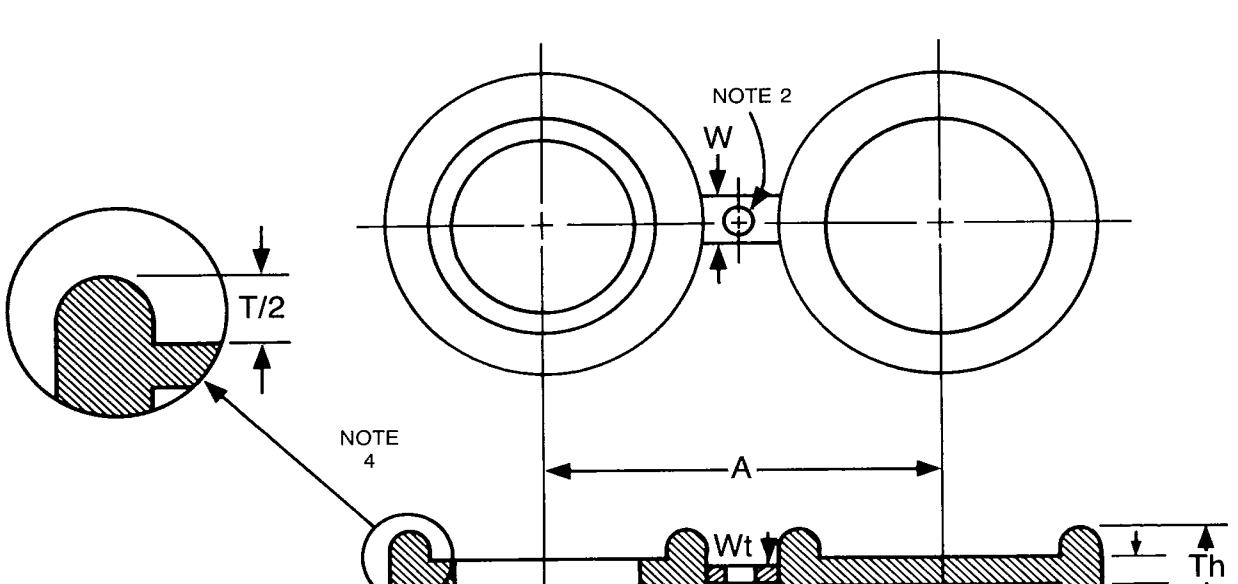
NPS	Inside Diameter B		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.62	67	0.25	6	1.50	38
¾	1.05	27	3.25	83	0.38	10	1.75	44
1	1.32	34	3.50	89	0.38	10	2.00	51
1 ¼	1.66	42	3.88	99	0.38	10	2.00	51
1 ½	1.90	48	4.50	114	0.38	10	2.25	57
2	2.38	60	5.00	127	0.50	13	2.25	57
2 ½	2.88	73	5.88	149	0.62	16	2.25	57
3	3.50	89	6.62	168	0.62	16	2.25	57
3 ½	4.00	102	7.25	184	0.75	19	2.50	64
4	4.50	114	7.88	200	0.62	16	2.50	64
5	5.56	141	9.25	235	0.75	19	3.00	76
6	6.62	168	10.62	270	0.88	22	3.25	83
8	8.62	219	13.00	330	1.00	25	3.75	95
10	10.75	273	15.25	387	1.12	29	4.00	102
12	12.75	324	17.75	451	1.38	35	4.75	121
14	14.00	356	20.25	514	1.50	38	5.00	127
16	16.00	406	22.50	572	1.62	41	5.00	127
18	18.00	457	24.75	629	1.75	44	5.00	127
20	20.00	508	27.00	686	2.00	51	5.00	127
24	24.00	610	32.00	813	2.25	57	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Oval ring joint dimensions shall be in accordance with ANSI B16.20 except $Th = T+t$ where *T* is the ring height specified in B16.20.

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Table 16—Dimensions of Class 600 Male Oval Ring-Joint Facing Figure 8 Blanks



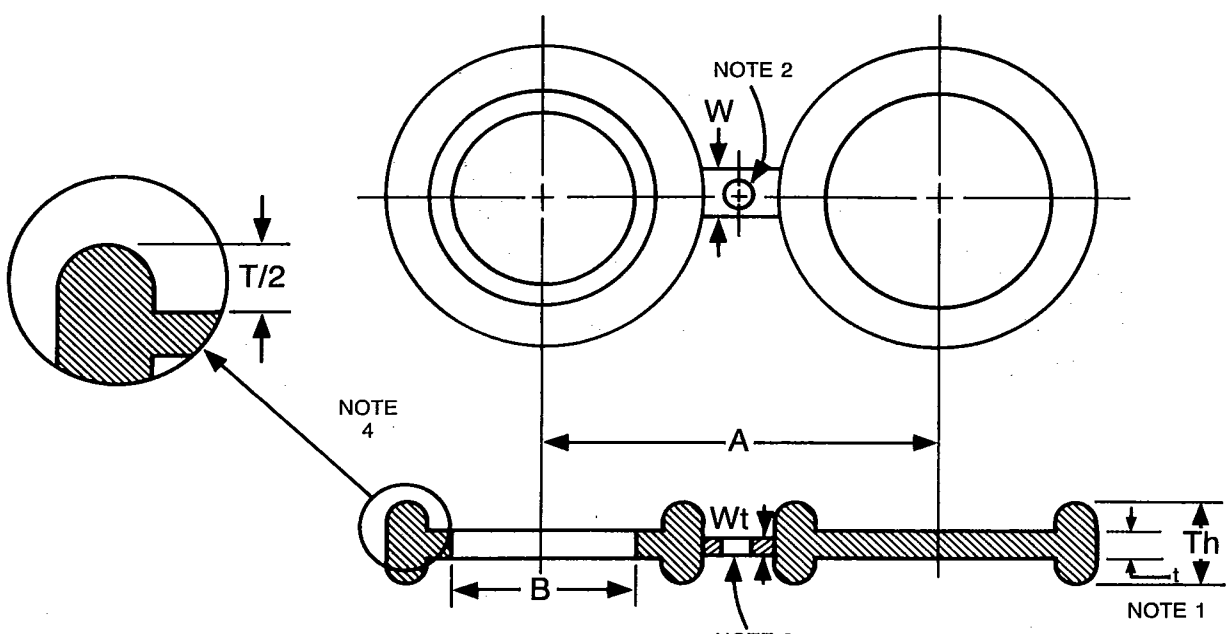
NPS	Inside Diameter *B*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	2.62	67	0.25	6	1.50	38
¾	1.05	27	3.25	83	0.38	10	1.75	44
1	1.32	34	3.50	89	0.38	10	2.00	51
1 ¼	1.66	42	3.88	99	0.38	10	2.25	57
1 ½	1.90	48	4.50	114	0.38	10	2.25	57
2	2.38	60	5.00	127	0.50	13	2.00	51
2 ½	2.88	73	5.88	149	0.62	16	2.25	57
3	3.50	89	6.62	168	0.62	16	2.62	67
3 ½	4.00	102	7.25	184	0.75	19	2.62	67
4	4.50	114	8.50	216	0.75	19	2.88	73
5	5.56	141	10.50	267	0.88	22	2.88	73
6	6.62	168	11.50	292	1.12	28	2.88	73
8	8.62	219	13.75	349	1.38	35	3.25	83
10	10.75	273	17.00	432	1.62	41	4.75	121
12	12.75	324	19.25	489	1.88	48	4.75	121
14	14.00	356	20.75	527	2.00	51	4.75	121
16	16.00	406	23.75	603	2.25	57	5.00	127
18	18.00	457	25.75	654	2.50	64	5.25	133
20	20.00	508	28.50	724	2.75	70	5.00	127
24	24.00	610	33.00	838	3.25	83	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Oval ring joint dimensions shall be in accordance with ANSI B16.20 except $Th = T+t$ where *T* is the ring height specified in B16.20.

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Table 17—Dimensions of Class 900 Male Oval Ring-Joint Facing Figure 8 Blanks



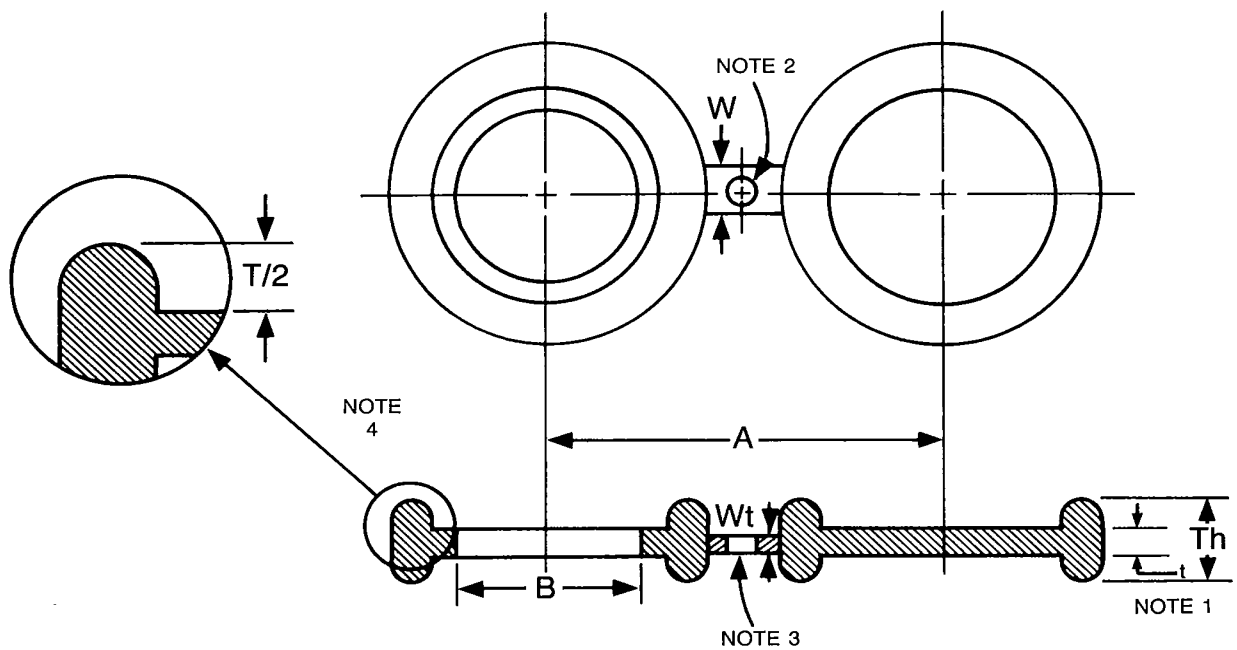
NPS	Inside Diameter B		Centerline Dimension A		Thickness t		Web Width W	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	3.25	83	0.38	10	1.50	38
¾	1.05	27	3.50	89	0.38	10	1.75	44
1	1.32	34	4.00	102	0.38	10	2.00	51
1 ¼	1.66	42	4.38	111	0.50	13	2.12	54
1 ½	1.90	48	4.88	124	0.50	13	2.12	54
2	2.38	60	6.50	165	0.62	16	2.00	51
2 ½	2.88	73	7.50	190	0.75	19	2.12	54
3	3.50	89	7.50	190	0.75	19	2.62	67
4	4.50	114	9.25	235	1.00	25	2.88	73
5	5.56	141	11.00	279	1.12	28	2.88	73
6	6.62	168	12.50	318	1.25	32	2.88	73
8	8.62	219	15.50	394	1.62	41	3.12	79
10	10.75	273	18.50	470	1.88	48	4.75	121
12	12.75	324	21.00	533	2.25	57	4.75	121
14	14.00	356	22.00	559	2.38	60	4.75	121
16	16.00	406	24.25	616	2.75	70	5.00	127
18	18.00	457	27.00	686	3.00	76	5.25	133
20	20.00	508	29.50	749	3.38	86	5.00	127
24	24.00	610	35.50	902	3.88	99	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Oval ring joint dimensions shall be in accordance with ANSI B16.20 except $Th = T+t$ where T is the ring height specified in B16.20.

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Table 18—Dimensions of Class 1500 Male Oval Ring-Joint Facing Figure 8 Blanks



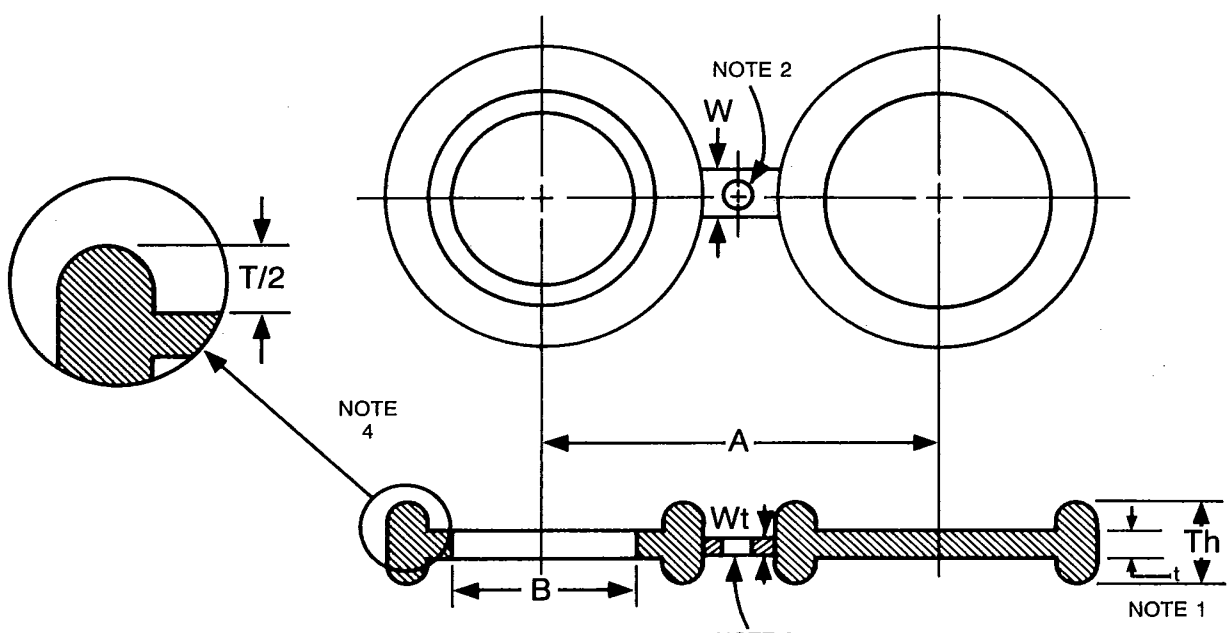
NPS	Inside Diameter *B*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	3.25	83	0.38	10	1.50	38
¾	1.05	27	3.50	89	0.38	10	1.75	44
1	1.32	34	4.00	102	0.50	13	2.12	54
1 ¼	1.66	42	4.38	111	0.50	13	2.12	54
1 ½	1.90	48	4.88	124	0.62	16	2.25	57
2	2.38	60	6.50	165	0.75	19	2.12	54
2 ½	2.88	73	7.50	190	0.88	22	2.25	57
3	3.50	89	8.00	203	1.12	28	2.88	73
4	4.50	114	9.50	241	1.25	32	3.00	76
5	5.56	141	11.50	292	1.50	38	3.00	76
6	6.62	168	12.50	318	1.62	41	3.12	79
8	8.62	219	15.50	394	2.00	51	3.38	86
10	10.75	273	19.00	483	2.50	64	5.25	133
12	12.75	324	22.50	572	2.88	73	5.25	133
14	14.00	356	25.00	635	3.12	79	5.50	140
16	16.00	406	27.75	705	3.50	89	5.75	146
18	18.00	457	30.50	775	3.88	99	6.00	152
20	20.00	508	32.75	832	4.25	108	6.50	165
24	24.00	610	39.00	991	5.00	127	7.00	178

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Oval ring joint dimensions shall be in accordance with ANSI B16.20 except $Th = T+t$ where T is the ring height specified in B16.20.

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Table 19—Dimensions of Class 2500 Male Oval Ring-Joint Facing Figure 8 Blanks



NPS	Inside Diameter *B*		Centerline Dimension *A*		Thickness *t*		Web Width *W*	
	inches	millimeters	inches	millimeters	inches	millimeters	inches	millimeters
½	0.84	21	3.50	89	0.50	13	1.62	41
¾	1.05	27	3.75	95	0.62	16	1.88	48
1	1.32	34	4.25	108	0.62	16	2.12	54
1 ¼	1.66	42	5.12	130	0.75	19	2.12	54
1 ½	1.90	48	5.75	146	0.88	22	2.38	60
2	2.38	60	6.75	171	1.00	25	2.25	57
2 ½	2.88	73	7.75	197	1.12	28	2.38	60
3	3.50	89	9.00	229	1.25	32	3.00	76
4	4.50	114	10.75	273	1.50	38	3.25	83
5	5.56	141	12.75	324	1.88	48	3.50	89
6	6.62	168	14.50	368	2.25	57	3.75	95
8	8.62	219	17.25	438	2.75	70	3.75	95
10	10.75	273	21.25	540	3.25	83	3.75	95
12	12.75	324	24.38	619	3.88	99	6.00	152

Notes:

1. Thickness (dimension *t*) includes a corrosion allowance of 0.05 inch (1.3 millimeters) for material groups 1.1, 1.7, 1.9, 1.10, and 1.12. Corrosion allowance is 0.00 inch for material groups 2.1, 2.2, 2.4, and 2.5.
2. Hole size (where required due to bolt spacing) shall be the same as the flange bolt hole.
3. The thickness of the web (or tie bar) dimension *Wt*, shall be 0.25 inch (6.4 millimeter) minimum, except when *t* is less than 0.25 inch, *Wt* shall equal *t*.
4. Oval ring joint dimensions shall be in accordance with ANSI B16.20 except $Th = T+t$ where T is the ring height specified in B16.20.

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